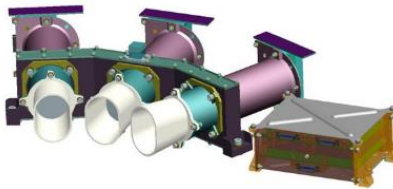


Chandrayaan-2 Terrain Mapping Camera-2 PDS4 Data Products and Archive Software Interface Specification



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Chandrayaan-2 Terrain Mapping Camera (TMC2)

PDS4 Data Archive

Software Interface Specifications

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1. Introduction

1.1 Purpose

The Data Products and Archive software interface specifications describes the data products, format and content of the Terrain Mapping Camera (TMC) - 2 Planetary Data System (PDS) data archive bundles in which data products with associated meta data, documentation, and supporting materials are stored. This document is intended to provide enough information to enable users to read and understand the TMC-2 data products stored in PDS.

1.2 Scope

The specifications in this document apply to all TMC-2 data products submitted for archival to the ISRO Science Data Archive (ISDA), for all the mission phases of the Chandrayaan-2.

1.3 Data Products and Archive SIS Content Overview

Section 2 of this document describes the Mission overview while Section 3 provides the characteristics of the terrain mapping camera instrument. Section 4 gives an overview of data organization and data flow while Section 5 describes data archive generation, delivery, and validation. Section 6 describes the archive structure and archive production responsibilities. Section 7 describes the file formats used in the archive, including the data product record structures. Individuals involved with generating the archive volumes are listed in Appendix-A While Appendix-B contains sample bundle label, Appendix-C, contains sample calibration collection label Appendix-D contains sample data product label, and Appendix-E contain visualization of PDS4 data products.

1.4 Audience

This document serves both as a SIS and Interface Control Document (ICD). It describes both the archiving procedure and responsibilities, and data archive conventions and format. It is designed to be used both by the instrument data processing team in generating the archive, and by those wishing to understand the format and content of the TMC2 PDS data product archive collection. Typically, these individuals would include scientists, data analysts, and software engineers.

1.5 Applicable Documents

- [1] Chandrayaan-2 Sequence of Events (ISRO-URSC-CHANDRAYAAN-2-PR-3171)
- [2] Chandrayaan-2 Configuration Summary Document (ISRO-URSC-CHANDRAYAAN-2-PR-3172)
- [3] Chandrayaan-2 Preliminary Design Review Data Products, Processing and Dissemination Plan - Volume 3A (Optical Payloads) (CHANDRAYAAN-2/DP/SAC/SIPG/HRDPD/TR-04/MAY 2017)
- [4] Incremental-Preliminary Design Review Document on Terrain Mapping Camera -2 (TMC-2), (SAC-SEDA-CH2-03-JUL14)
- [5] Chandrayaan-2 Science data management and Archive plan (Chandryaan-2/DP/SAC/SIPG/HRDPD/TR-06/July 2018)
- [6] Chandrayaan-2 SAC DP – ISSDC Interface for Archival (Chandrayaan-2/DP/SAC/SIPG/HRDPD/TR-13/Dec. 2018)
- [7] PDS4 Concepts Data Design Working Group October 1, 2018 version 1.11.0
- [8] Planetary Data System Standards Reference Version 1.11.0 October 1, 2018, Jet Propulsion Laboratory
- [9] The PDS4 Data Provider's Handbook Guide to Archiving Planetary Data Using the PDS4 Standard Version 1.11.0 October 1, 2018
- [10] PDS4 Data Dictionary Abridged – Version 1.11.0.0
- [11] List of Supplemental Formats (Approved by PDS Management Council: 2017-08-30)

1.6 Abbreviations Acronyms, and Description

Following **Table** brings out the list of definitions, acronyms & abbreviations used in this document.

Table 1 Acronym, Abbreviations and Description

Acronym	Abbreviations/Description
PDS	Planetary Data System
ISRO	Indian Space Research Organization
ISSDC	Indian Space Data Centre
ISDA	ISRO Science Data archive
SAC	Space Applications Centre
DP	Data Processing
TMC2	Terrain Mapping Camera -2
SPICE	Spacecraft Planet Instrument Camera Event
ICD	Interface Control Document
Kernel	Primary SPICE data sets
DPD	Deputy Project Director
Archive Team	Collection of teams from science (PI), Payload, Data Processing and ISSDC Operations

1.7 Glossary

Table 2 shows glossary of terms used in PDS4 domain.

Table 2 PDS4 Glossary

Terms	Definition
Archive	A place in which public records or historical documents are preserved; also the material preserved – often used in plural. The term may be capitalized when referring to all of PDS holdings – the PDS Archive.
Basic Product	The simplest product in PDS4; one or more data objects (and their description objects), which constitute (typically) a single observation, document, etc. The only PDS4 products that are not basic products are collection and bundle products
Bundle Product	A list of related collections. For example, a bundle could list a collection of raw data obtained by an instrument during its mission lifetime, a collection of the calibration products associated with the instrument, and a collection of all documentation relevant to the first two collections.
Class	The set of attributes (including a name and identifier) which describes an item defined in the PDS Information Model. A class is

Terms	Definition
	generic – a template from which individual items may be constructed.
Collection Product	A list of closely related basic products of a single type (e.g. observational data, browse, documents, etc.). A collection is itself a product (because it is simply a list, with its label), but it is not a basic product.
Data Object	A generic term for an object that is described by a description object. Data objects include both digital and non-digital objects.
Description Object	An object that describes another object. As appropriate, it will have structural and descriptive components. In PDS4 a 'description object' is a digital object – a string of bits with a predefined structure.
Digital Object	An object which consists of real electronically stored (digital) data.
Identifier	A unique character string by which a product, object, or other entity may be identified and located. Identifiers can be global, in which case they are unique across all of PDS (and its federation partners). A local identifier must be unique within a label.
Label	The aggregation of one or more description objects such that the aggregation describes a single PDS product. In the PDS4 implementation, labels are constructed using XML.<say what it contains>
Logical Identifier (LID)	An identifier which identifies the set of all versions of a product.
Versioned Logical Identifier (LIDVID)	The concatenation of a logical identifier with a version identifier, providing a unique identifier for each version of product.
Manifest	A list of contents.
Metadata	Data about data – for example, a 'description object' contains information (metadata) about an 'object.'
Non-Digital Object	An object which does not consist of digital data. Non-digital objects include both physical objects like instruments, spacecraft, and planets, and non-physical objects like missions, and institutions. Non-digital objects are labeled in PDS in order to define a unique identifier (LID) by which they may be referenced across the system.
Object	A single instance of a class defined in the PDS Information Model.
PDS Information Model	The set of rules governing the structure and content of PDS metadata. While the Information Model (IM) has been implemented in XML for PDS4, the model itself is implementation independent.
Product	One or more tagged objects (digital, non-digital, or both) grouped together and having a single PDS-unique identifier. In the PDS4

Terms	Definition
	implementation, the descriptions are combined into a single XML label. Although it may be possible to locate individual objects within PDS (and to find specific bit strings within digital objects), PDS4 defines 'products' to be the smallest granular unit of addressable data within its complete holdings.
Tagged Object	An entity categorized by the PDS Information Model, and described by a PDS label.
Registry	A data base that provides services for sharing content and metadata.
Repository	A place, room, or container where something is deposited or stored (often for safety).
XML	eXtensible Markup Language.
XML schema	The definition of an XML document, specifying required and optional XML elements, their order, and parent-child relationships.

2. Mission Overview

Chandrayaan-2, the second Indian mission to the moon was launched on 22nd July 2019 with Orbiter, Lander and Rover configuration. The mission is a highly complex mission, which represents a significant technological leap compared to the previous missions of ISRO. It comprised an Orbiter, Lander and Rover to explore the unexplored South Pole of the Moon. The mission is designed to expand the lunar scientific knowledge through detailed study of topography, seismography, mineral identification and distribution, surface chemical composition, thermo-physical characteristics of top soil and composition of the tenuous lunar atmosphere, leading to a new understanding of the origin and evolution of the Moon. After the injection of Chandrayaan-2, a series of maneuvers were carried out to raise its orbit and on August 14, 2019, following Trans Lunar Insertion (TLI) maneuver, the spacecraft escaped from orbiting the earth and followed a path that took it to the vicinity of the Moon. On August 20, 2019, Chandrayaan-2 was successfully inserted into lunar orbit. While orbiting the moon in a 100 km lunar polar orbit, on September 02, 2019, Vikram Lander was separated from the Orbiter in preparation for landing. Subsequently, two de-orbit maneuvers were performed on Vikram Lander so as to change its orbit and begin circling the moon in a 100 km x 35 km orbit. Vikram Lander descent was as planned and normal performance was observed up to an altitude of 2.1 km. Subsequently communication from lander to the ground stations was lost. The Orbiter placed in its intended orbit around the Moon will enrich our understanding of the moon's evolution and mapping of the minerals and water molecules in Polar regions, using its eight state-of-the-art scientific instruments. The Orbiter camera is the highest resolution camera (0.3 m) in any lunar mission so far and will provide high resolution images which will be immensely useful to the global scientific community. The precise launch and mission management has ensured a long life of almost seven years instead of the planned one year.

2.1 Mission Objective

To expand the lunar scientific knowledge through detailed study of topography, mineralogy, surface chemical composition, thermo-physical characteristics and tenuous lunar atmosphere leading to a better understanding of the origin and evolution of the Moon

2.2 Payload Objective

Primary objective is mapping the lunar surface in the panchromatic spectral band (0.5-0.8 microns) with a high spatial resolution of 5 m and a swath of 20 km from 100 km lunar polar orbit

3. Instrument Overview

Terrain Mapping Camera-2 (TMC-2), on-board Chandrayan-2 Orbiter-craft is a follow-on of the TMC on-board Chandrayan-1 TMC-2 payload is configured to provide panchromatic images (0.4 μ m to 0.85 μ m) in 5m spatial resolution & stereo triplets (fore, nadir and aft views) from 100 km circular orbit around moon for preparing detailed 3-D map or Digital Elevation model (DEM) of the complete lunar surface (especially the smaller objects like craters, rilles, flow structures, smaller graben forms). DEM derived from TMC-2 will be used for detail morphometric analysis of lunar surface.

TMC-2 had same system specifications of TMC viz. IGFOV of 5m $\times$ 5m, swath coverage of 20km, 12bit digitization etc. except for spacecraft resource requirements.

Prime challenge in development of TMC-2 is miniaturization keeping mass <4kg. Table 3 shows the payload features.

Table 3 TMC2 Payload Features

S. No.	Parameter	Value
1	Altitude	100km
2	Spatial Sampling	Along track 5m Across track 5m
3	Swath	20km
4	Stereo mode	3-view along track, B/H=1
5	Spectral range	Panchromatic 0.5-0.85 μ m
6	Quantization	12 bit
7	Data rate	16.2 Mbps/view
8	Gain/Exposure	4-Exposures and single gain
9	MTF	>10% @72 lp/mm
10	SNR (goal)	~75@1.2mW/cm ² /str/ μ m >500@14mW/cm ² /str/ μ m
11	Power	<3.9W (Raw)
12	Weight	<3.5kg (Goal)
13	EOM (YxPxR)	280mm x 120mm x 440mm

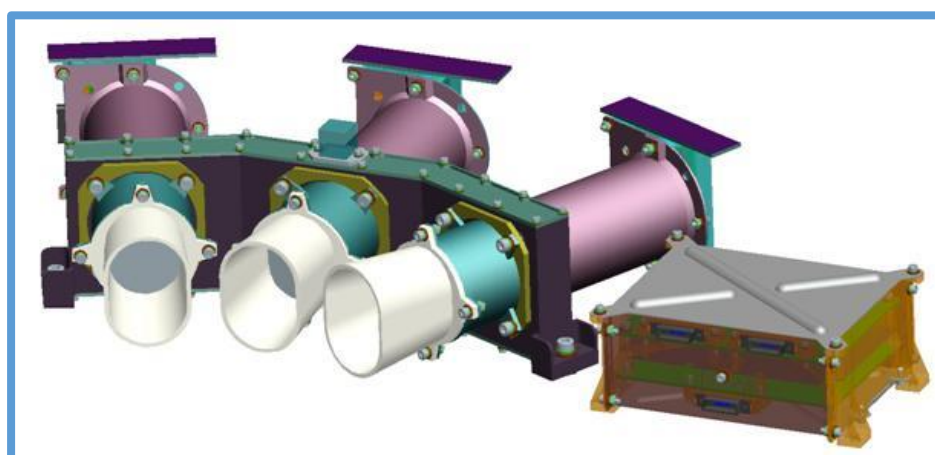
Based on TMC experience, single gain (G1) is found suitable for covering signal range. However, 4 exposure settings are retained for fine tuning. **Table 4** shows the expected signal for the extreme conditions.

Table 4 TMC-2 Signal Range

Scene and illumination	Radiance (mW/cm ² /str/μm)	Signal* (ke-/pixel)	Signal (relative)
Mature mare soil at 89 deg. Lat. (for Θ sun=89.13 deg.	0.0427	1.1	1
Mature mare soil at 89 deg. Lat. (for Θ sun=0 deg.	2.81	71	65
Fresh rock surface at 0 deg. Lat. (for Θ sun=0 deg.	14.2	360	330
* For: Ti=3.236ms, BW=0.5-0.85μm,			

3.1 Camera Configuration

Figure 1 shows the TMC2 isometric view of EOM and CE.

**Figure 1 Isometric view of EOM and CE**

3.2 Optics

As the ratio of ground resolution to altitude of TMC-2 remains same as TMC-1, optics specification remains same for both. **Table 5** provides the specification for TMC-2 Optics.

Table 5 TMC-2 Optics Specifications

Parameters	Specifications
Spectral bands	Panchromatic (0.50 - 0.85 μm)
Focal length	140mm
F/no.	F/4
Field Of View (FOV) Swath direction Along track/Look	± 5.72° + 25°, 0°, -25°

angles	
Nyquist Spatial frequency	72 lp/mm

3.3 Principal of Operations

For stereoscopic imaging, TMC-2 has three telescopes with 4000 one dimensional detectors looking in fore (+25°), nadir (0°) and aft (-25°) directions respectively. **Figure-2** shows the schematic of principle of operations of TMC-2.

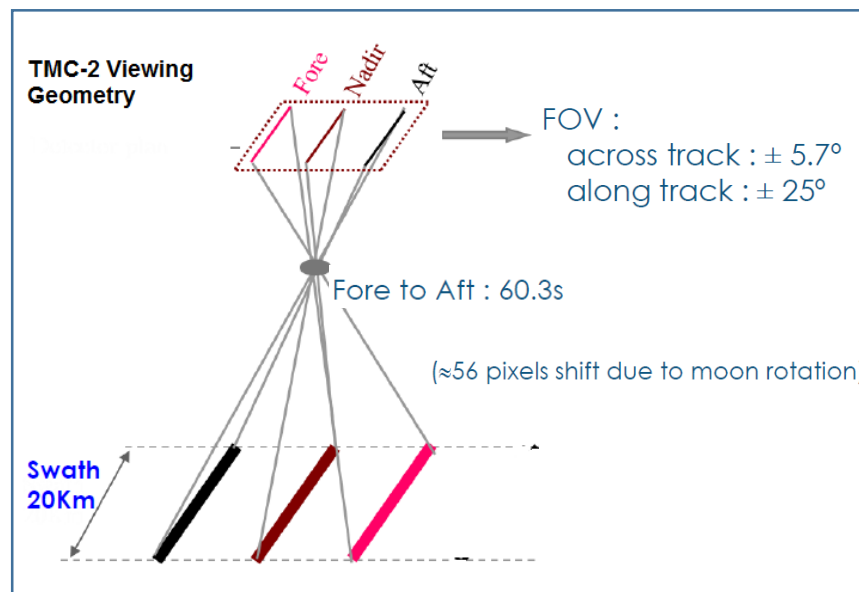


Figure 2 TMC-2 Principle of Operations

The DEM generation process involves Image matching between the suitable two or three images of the triplet, Lunar Control Point (LCP) identification, resection, intersection, irregular DEM points generation and regular DEM and corresponding Ortho-Image generation using Interpolation.

4. Data Overview

This section provides a high level description of archive organization under the PDS4 Information Model (IM) as well as the flow of the data from the spacecraft through delivery to PDS. Unless specified elsewhere in this document, the Chandrayaan-2 TMC-2 archive confirms with version 1.7.0.0 of the PDS4 IM [4] and version 1.7 of the Chandrayaan-2 mission schema. A list of the XML Schema and Schematron documents associated with this archive are provided in **Table 6** below.

Table 6 Chandrayaan-2 TMC2 Archive Schema and Schematron

XML Document	Steward	Product LID
PDS Master Schema, v.1.0.0.0	ISDA	urn:isro:isda:system_bundle:xml_schema:pds-xml_schema

4.1 Data Processing Levels

A number of different systems may be used to describe data processing level. This document refers to data by their PDS4 reduction level. Three different types of data products are envisaged for TMC-2 payload of Chandrayaan-2. **Table-7** provides a description of these levels along with the equivalent designations used in other systems.

Table 7 Chandrayaan-2 TMC2 Data Processing Levels Mapping with PDS4 Level

PDS4 Level	PDS4 Level Description	ISRO Level	CODMAC Level
Raw	Original data from an instrument. If compression, reformatting, packetization, or other translation has been applied to facilitate data transmission or storage, those processes will be reversed so that the archived data are in a PDS approved archive format.	0	2
Calibrated	Data converted to physical units, which makes values independent of the instrument.	1	3
Derived	Results that have been distilled from one or more calibrated data products (for example, maps, gravity or magnetic fields, or ring particle size distributions). Supplementary	2	4+

	data, such as calibration tables or tables of viewing geometry, used to interpret observational data should also be classified as 'derived' data if not easily matched to one of the other three categories.		
--	--	--	--

4.2 Data Products

A PDS product consists of one or more digital and/or non-digital objects, and an accompanying PDS label file. Labeled digital objects are data products (i.e. electronically stored files). Labeled non-digital objects are physical and conceptual entities which have been described by a PDS label. PDS labels provide identification and description information for labeled objects. The PDS label defines a Logical Identifier (LID) by which any PDS labeled product is referenced throughout the system. In PDS4 labels are XML formatted ASCII files. More information on the formatting of PDS labels is provided in Section 6.3. More information on the usage of LIDs and the formation of Chandrayaan-2 LIDs is provided in Section 6.1. The PDS4 Data Products for TMC2 defined based on the defined data processing levels (Level-0, Level-1 and Level-2) as Raw, Calibrated and Derived data products. Table 8 shows the data products definitions for each levels of data processing. To visualize the data products please refer Annexure E. more details shown in next sub sections.

Table 8 Chandrayaan-2 TMC2 Data Products Definitions

Data Products Levels	Level-0	Level-1	Level-2
Data Products Definition	Raw data without any data processing correction along with system level corner coordinates	Radiometrically corrected, refined corner coordinates tagged and grid data	Radiometrically corrected & geometrically corrected data also known as higher level products - DEM (10m) and Ortho image (5m)
PDS4 Level	Raw	Calibrated	Derived

4.2.1 Raw Data Products

For each payload observation, PDS4 data products are generated (based on product id & sensor id derived from file name (section 6.5)), zipped and organized under year month day wise collection directory defined under instrument collection structure shown in section 6.4. For example, user has downloaded the zip product as

- ch2_tmc_nrf_20191015T102125157610840_d_img_blr.zip
- ch2_tmc_nra_20191015T102125154360840_d_img_blr.zip
- ch2_tmc_nrn_20191015T102125154360840_d_img_blr.zip

Figure 3 shows unzipped structure of TMC2 Raw Data Product zip file at user end.

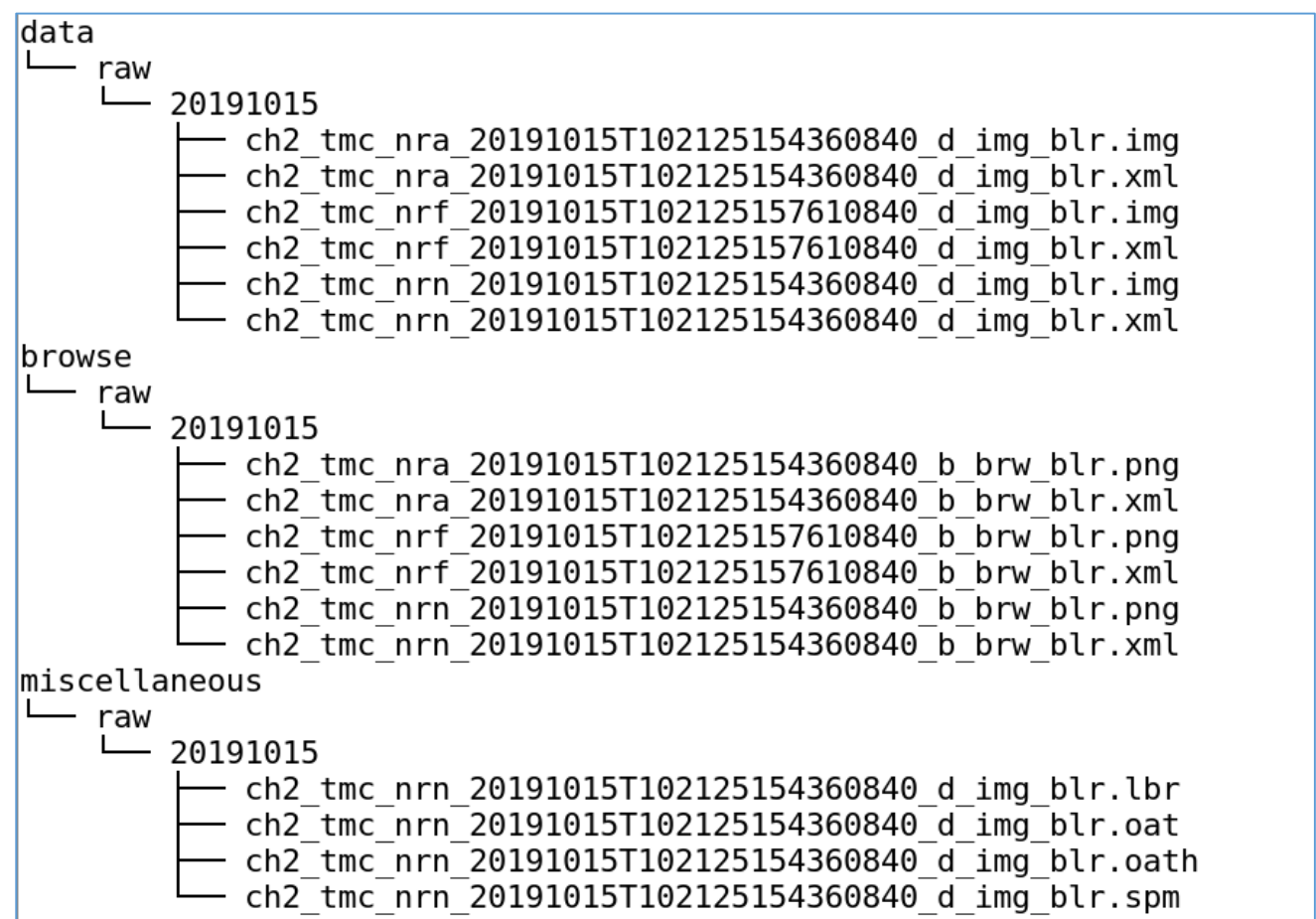


Figure 3 TMC2 Raw Data Archive

Raw Data Products contains raw data without any correction along with the system level corner coordinates. Table 9 shows the details of individual data products (with the data formats) present inside the zip product.

Table 9 TMC2 Raw Data Products

PDS4 Data Product Levels	PDS4 Data Products	Data Formats	Description	Visualization tools available
For sensor – Fore Aft Nadir Raw	Data Raw Image	Binary	The PDS Raw data in generic binary	Any binary image viewer – imageJ, envi, erdas Provide image width, image height and data type from label file
	Data Raw Label	XML	The PDS Label Product contains description about the Raw Product.	PDS4 Viewer
	Browse Raw Image	PNG	Sub sampled browse raw image	Any image viewer can be used to load the png file, so no need to load label file using PDS4 Viewer
	Browse Raw Label	XML	The PDS Label Product contains description about the Browse Raw Product.	
	Miscellaneous Data (non PDS data)	Text	Contains four files Orbit & Attitude (oat) Orbit & Attitude header(oath) Liberation angle (lbr) Sun parameter file spm	Any text viewer software will be able to view the data

4.2.2 Calibrated Data Products

For each payload observation, PDS4 data products are generated (based on product id derived from file name (section 6.5)), zipped and organized under year month day wise collection directory defined under instrument collection structure shown in section 6.4. For example, user has downloaded the zip product as

- ch2_tmc_ncf_20191015T102125157610840_d_img_blr.zip
- ch2_tmc_nca_20191015T102125154360840_d_img_blr.zip
- ch2_tmc_ncn_20191015T102125154360840_d_img_blr.zip

Figure 4 shows unzipped structure of OHRC Calibrated Data Product zip file at user end.

```

data
├── calibrated
│   └── 20191015
│       ├── ch2_tmc_nca_20191015T102125154360840_d_img_blr.img
│       ├── ch2_tmc_nca_20191015T102125154360840_d_img_blr.xml
│       ├── ch2_tmc_ncf_20191015T102125157610840_d_img_blr.img
│       ├── ch2_tmc_ncf_20191015T102125157610840_d_img_blr.xml
│       ├── ch2_tmc_ncn_20191015T102125154360840_d_img_blr.img
│       └── ch2_tmc_ncn_20191015T102125154360840_d_img_blr.xml
├── browse
│   └── calibrated
│       └── 20191015
│           ├── ch2_tmc_nca_20191015T102125154360840_b_brw_blr.png
│           ├── ch2_tmc_nca_20191015T102125154360840_b_brw_blr.xml
│           ├── ch2_tmc_ncf_20191015T102125157610840_b_brw_blr.png
│           ├── ch2_tmc_ncf_20191015T102125157610840_b_brw_blr.xml
│           ├── ch2_tmc_ncn_20191015T102125154360840_b_brw_blr.png
│           └── ch2_tmc_ncn_20191015T102125154360840_b_brw_blr.xml
└── miscellaneous
    └── raw
        └── 20191015
            ├── ch2_tmc_ncn_20191015T102125154360840_d_img_blr.lbr
            ├── ch2_tmc_ncn_20191015T102125154360840_d_img_blr.oat
            ├── ch2_tmc_ncn_20191015T102125154360840_d_img_blr.oath
            └── ch2_tmc_ncn_20191015T102125154360840_d_img_blr.spm

```

Figure 4 TMC2 Calibrated Data Archive

Calibrated Data Products contains radiometrically corrected data along with the refined corner coordinates. Table 10 shows the details of individual data products (with the data formats) present inside the zip product.

Table 10 TMC2 Calibrated Data Products

PDS4 Data Product Levels	PDS4 Data Products Types	Data Formats	Description	Visualization tools available
For sensor – Fore Aft Nadir Calibrated	Data Calibrated Image	Binary	The PDS Radio metrically corrected data in generic binary format	Any binary image viewer – imageJ, envi, erdas Provide image width, image height and data type from label file
	Data Calibrated Label	XML	The PDS Label Product contains description about the Calibrated Product.	PDS4 Viewer
	Browse Calibrated Image	PNG	Sub sampled browse calibrated image	Any image viewer can be used to load the png file, so no

PDS4 Data Product Levels	PDS4 Data Products Types	Data Formats	Description	Visualization tools available
	Browse Calibrated Label	XML	The PDS Label Product contains description about the Browse Calibrated Product.	need to load label file using PDS4 Viewer
	Geometric Calibrated Grid	CSV	A grid file containing lon, lat, scan and pix as four columns along with the data	Any text editor, ms excel can be used to load the csv file.
	Geometric Calibrated Grid Label	XML	The PDS Label Product contains description about the Geometry Calibrated Product.	PDS4 Viewer
	Miscellaneous Data (non PDS data)	Text	Contains four files Orbit & Attitude (oat) Orbit & Attitude header(oath) Liberation angle (lbr) Sun parameter file spm	Any text viewer software will be able to view the data

4.2.3 Derived Data Products

For each payload observation, PDS4 data products are generated (based on product id derived from file name (Section 6.5)), zipped and organized under year month day wise collection directory defined under instrument collection structure shown in section 6.4. For example, user has downloaded the zip product as

- ch2_tmc_ndn_20191015T102125154360840_d_dtm_blr.zip
- ch2_tmc_ndn_20191015T102125154360840_d_oth_blr.zip

Figure 5 shows unzipped structure of TMC2 Calibrated Data Product zip file at user end.

```

data
├── derived
│   └── 20191015
│       ├── ch2_tmc_ndn_20191015T102125154360840_d_dtm_blr.tif
│       ├── ch2_tmc_ndn_20191015T102125154360840_d_dtm_blr.xml
│       ├── ch2_tmc_ndn_20191015T102125154360840_d_oth_blr.tif
│       └── ch2_tmc_ndn_20191015T102125154360840_d_oth_blr.xml
└── browse
    ├── derived
    │   └── 20191015
    │       ├── ch2_tmc_ndn_20191015T102125154360840_b_bdt_blr.png
    │       ├── ch2_tmc_ndn_20191015T102125154360840_b_bdt_blr.xml
    │       ├── ch2_tmc_ndn_20191015T102125154360840_b_bot_blr.png
    │       └── ch2_tmc_ndn_20191015T102125154360840_b_bot_blr.xml

```

Figure 5 TMC2 Derived Data Archive

Derived Data Products contains radiometrically and geometrically corrected data along with the corner coordinates. Table 11 shows the details of individual data products (with the data formats) present inside the zip product.

Table 11 TMC2 Derived Data Products

PDS4 Data Product Levels	PDS4 Data Products Types	Data Formats	Description	Visualization tools available
Derived	Data DEM Image	Geo Tiff	DEM image data	Any Tiff image viewer – imageJ, envi, erdas
	Data DEM Label	XML	The PDS Label Product contains description about the Derived DEM data Product.	
	Data Ortho Image	Geo Tiff	Ortho rectified image data	Any Tiff image viewer – imageJ, envi, erdas
	Data Ortho Label	XML	The PDS Label Product contains description about the Derived Ortho Product.	
	Browse DEM Image	PNG	Sub sampled browse DEM image	Any image viewer can be used to load the png file, so no need to load label file using PDS4 Viewer
	Browse DEM Label	XML	The PDS Label Product contains description about the Browse Derived DEM data Product.	
	Browse Ortho Image	PNG	Sub sampled browse ortho image	Any image viewer can be used to load the

	Browse Ortho Label	XML	The PDS Label Product contains description about the Browse Derived Ortho Product.	png file, so no need to load label file using PDS4 Viewer
--	--------------------	-----	--	---

DEM (Digital Elevation model) derived from TMC will also be used for photometric correction of Hyper-spectral data (obtained from IIRS Instrument). The two extreme targets – fresh rock surface and mature mare soil, along with the factors of seasonal variation, lat-long of the scene and anisotropic reflectance of lunar surface, represent the dynamic range of the signal.

4.3 Data Products Bundle

The TMC2 data archive is organized into 3 bundles at data level. A description of each bundle is provided in Table 12. A more detailed description of the contents and format of each bundle is provided in Section 6.2.

Table 12 TMC2 Data Products Bundles

Bundle Logical Identifier	PDS4 Reduction Level	Description	Data Provider
urn:isro:isd a:ch2.tmc.raw	Raw	Original data from an instrument. If compression, reformatting, packetization, or other translation has been applied to facilitate data transmission or storage, those processes will be reversed so that the archived data are in a PDS approved archive format	ISSDC
urn:isro:isd a:ch2.tmc.calibrated	Calibrated	Data converted to physical units, which makes values independent of the instrument.	ISSDC
urn:isro:isd a:ch2.tmc.derived	Derived	Results that have been distilled from one or more calibrated data products (for example, maps, gravity or magnetic fields, or ring particle size distributions). Supplementary data, such as calibration tables or tables of viewing geometry, used to interpret observational data should also be classified as 'derived' data if not easily matched to one of the other three categories	ISSDC

4.4 Data Flow

This section describes only those portions of the Chandrayaan-2 data flow at ISSDC that are directly connected to archiving. The ISSDC will maintain an active archive of all Chandrayaan-2 science data, and will provide the PI with direct access through the life of the Chandrayaan-2 mission. After the end of the Chandrayaan-2 project, ISDA will be the sole long-term archive for all public Chandrayaan-2 data.

5. Archive Generation

The TMC2 archive products are produced by the SAC DP team in cooperation with the ISSDC. The archive volume creation process will be described in this section. Archived data established at ISSDC from the SAC DP software running at ISSDC, are made available to users electronically as soon as it qualifies as a science product but not later than nine months after the delivery and validation of the data.

5.1 Data Processing and Production Software

After receiving Level 0 data from the level-0 software at ISSDC, the ISSDC will process the Level 0 data into science data products using software provided by the SAC DP. The ISSDC will provide the Level 0 data files with all the ancillary data required for science processing to the SAC DP Software. From this data, the SAC DP Software will generate Level 0/1/2 raw, calibrated and derived science data products. The science data products that the SAC DP Software delivers to the ISSDC will be stored by the ISSDC for the duration of the project, and will be made available to the PI team or as per the ISRO data distribution policy. The ISSDC will deliver archival-quality science data products to the ISDA for distribution to the public and long-term archiving in accordance with the TMC-2 Data Products SIS & Archive (this document).

5.2 Data Validation

There are two types of data validation. The first is validation of the science data and the second one is validation of the compliance of the archive with PDS archiving and distribution requirements. The first type of validation is carried out by the PI Team, and the second is overseen by the ISDA, in coordination with the PI Team. The formal validation of data content, adequacy of documentation, and adherence to PDS archiving and distribution requirements is scheduled and coordinated by the ISDA. This review may include peer review from the science community, at the ISDA's discretion. The review process may result in "liens," which are actions that are recommended by the reviewers or by PDS personnel to correct the archive. All liens must be resolved by the data-set provider. Once the liens are cleared, the ISDA completes a final validation before release.

5.3 Data Transfer Methods and Delivery Schedule

The Data Processing & Archival Software will be running at ISSDC during each mission phases of Chandrayaan-2. The products will be generated at ISSDC. Initially active archive will be established data will be collected over six months. The products will be available for the PI as initial validation of data. After six months, Long Term Archive will be established during next six months. Table 13 shows TMC data release schedule plan for Chandrayaan-2 mission.

Table 13 Data Release Schedule Plan

Payload	Release*	Data Collection Period (from date to date)	Lock-in-period (as per SSPO ISRO HQ)**	Date Released to Public***	Products
TMC 2	Release 1 (first imaging season of 4 months)	25 Nov. 2019 to 12 Feb 2020	9 months	12 Nov. 2020	Raw, Calibrated and derived data sets
	Release 2 (second imaging season)	May 2020 to Sept. 2020	9 months	Oct. 2021	Raw, Calibrated and derived data sets
	Release 3 (third imaging season)	Nov. 2020 to April 2021	9 months	Oct. 2021	Raw, Calibrated and derived data sets
	Release 4 (fourth imaging season)	May. 2021 to August 2021	6 months	Feb 2022	Raw, Calibrated and derived data sets
	Release 5 (fifth and sixth imaging season)	Season 5 Nov. 2021 to Feb. 2022 Season 6 May 2022 to August 2022	6 months	Sep. 2022	Raw, Calibrated and derived data sets
	Release 6 (seventh imaging season)	Season 7 Nov. 2022 to Feb. 2023	6 month	March 2023	Raw, Calibrated and derived data sets
	Release 7 (eighth imaging season)	Season 8 April 2023 to July. 2023	7 month	October 2023	Raw, Calibrated and derived data sets
	Release 8 (ninth imaging season)	Season 9 June 2023 to Feb. 2024	1 month	April 2024	Raw, Calibrated and derived data sets
	Release 9 (tenth imaging season)	Season 10 Feb 2024 to August 2024	1 month	August 2024	Raw, Calibrated and derived data sets
	Release 10 (eleventh imaging season)	Season 11 Oct 2024 to Feb 2025	1 month	Feb 2025	Raw, Calibrated and derived data sets
	Release 11 (12th imaging season)	Season 12 April 2025 to May 2025	1 month	June 2025	Raw and Calibrated and data sets
	Release 12 (13th imaging season)	Season 13 Nov 2025 to Feb 2026	1 month	Nov 2025	Raw and Calibrated and data sets
	Release 13 (14th imaging season)	Season 14 May 2026 to Aug 2026	1 month	July 2026	Raw and Calibrated and data sets

- * Archive for the duration of 1 months (Nov. to Nov) will be released to the user.
- ** The lock in period duration will be required for initial data verification and validation and finally peer review process of the PDS data sets which will be going to release.
- ** Release dates are dates that the ISDA releases Chandrayaan-2 archives to the public. Chandrayaan-2 delivers archives to the ISDA at least two weeks before the date of release to the public.

5.4 Data Product and Archive Volume Size Estimates

The total Archive Volume Size is worked out based on the data size, no. of orbits per day, mission life.

5.5 Backups and duplicates

The ISDA is responsible for maintaining at least three copies of its science archives and for delivering one copy of the data to the ISSDC for archive. As archives are released, ISSDC generate at least three copies on appropriate physical media for long-term storage.

6. Archive Organization and File Naming Conventions

6.1 Namespace Registration

PDS4 archive standard is adopted for Chandrayaan-2 mission. PDS4 is international archive standard being adopted by various space agencies across globe. There is International Planetary Data Alliance (IPDA), where all space agencies are members, ISRO is also one of them. IPDA has adopted PDS4. First step towards starting archive activities for any space agency is to register namespace via IPDA. ISRO has submitted namespace to IPDA. The namespace will be used as root to the archive of ISRO's upcoming and future planetary missions. For Chandrayaan-2 mission following name space will be used which is shown in the Table 15.

Table 14 ISDA Namespace for PDS4

Namespace ID	Logical Identifier prefix	Authority	Steward	Steward_ID	Provider/contact Point
isda	urn:isro:isda	ISRO Science Data Archive	ISRO	isda	Ajay Kumar Prashar T P Srinivasan B.N. Ramakrishna

6.2 Logical Identifiers

Logical Identifier(LID) is unique to the product. Versioned Identifier (VID) indicates the versions improvements with respect to the products. All LIDs consist of a series of colon-separated segments.

Namespace:{BundleId}_{SubBundleId}:[{CollectionId}]_PDS4DataProductLevel:[{ProductId}]]. Table 15 shows values for LID construction.

Table 15 LID for Chandrayaan-2 mission

Namespace	Bundle Id	Sub Bundle Id	Collection Id	PDS4 Data Product Level	Product Id
urn:isro:isda	ch2	cho	tmc	raw calibrated derived	Address Data Products SIS document or EAICD

The following table 17 shows the construction of LID to be used for all the scientific payloads of Chandrayaan-2 mission.

Table 16 LID for Chandrayaan-2 mission instruments

Instrument id	Logical Identifier
tmc	urn:isro:isda:ch2_cho.tmc:data_raw:productId urn:isro:isda:ch2_cho.tmc:data_calibrated:productId urn:isro:isda:ch2_cho.tmc:data_derived:productId

Products, collections, and bundles may evolve over time as calibration or other processing changes lead to improved versions. PDS appends a VID (version Identifiers) of the form M.n to LID to indicate different versions. The suffix is separated from LID by double colon Archive structure is designed both at mission phase level and instrument level.

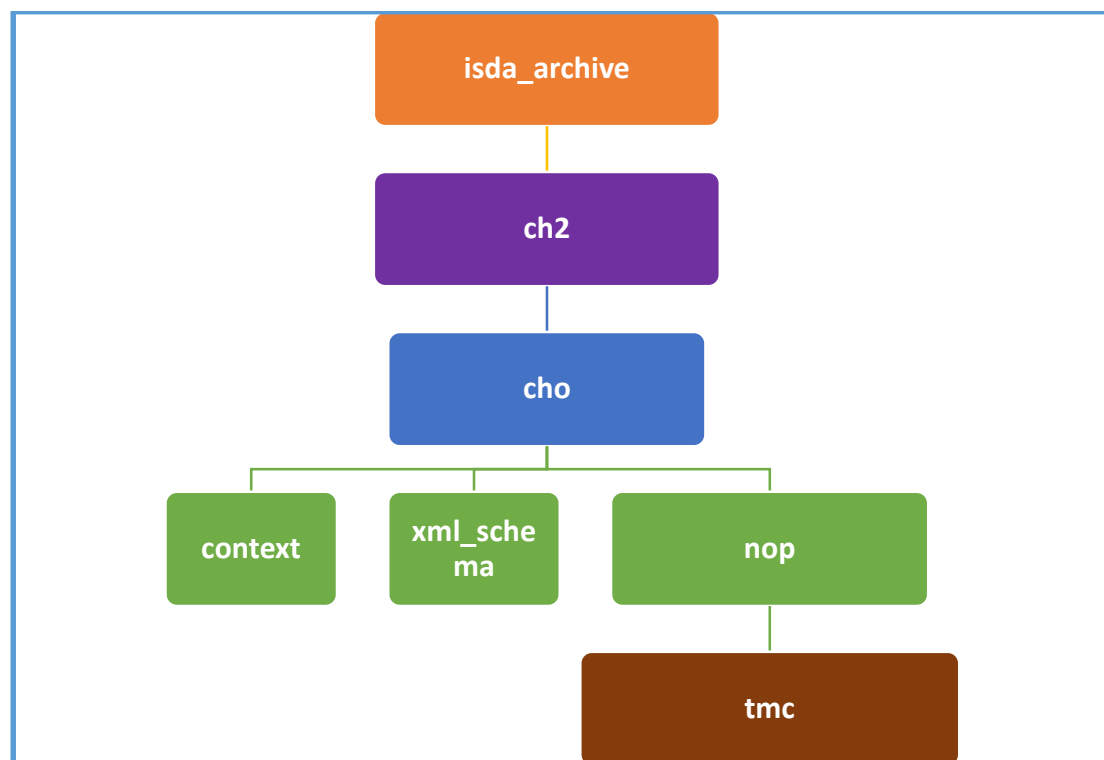
6.3 Mission Archive Structure at Bundle Level

At mission level, data is organized as mission id wise bundle. The mission id wise bundle at next level is organized based on the instrument host id wise bundle. The instrument host id bundle at next level is organized based on the mission phase id. Each mission phase id is having instrument id wise collection. All these mission attributes - mission id, instrument host id, mission phase id and instrument id are defined under archive conventions. For TMC2 instrument, table 18 shows values defined for mission attributes in archive conventions.

Table 17 Mission Attributes defined at structure level

Mission Attributes	Values	Defined at structure level
mission id	ch2	ch2_bundle
instrument host id	cho	cho_bundle
mission phase id	nop	nop (normal operation phase)
instrument id	tmc	tmc_collection

All these attributes mentioned in above table are used in designing the archive structure at mission level. The figure 7 shows the mission level data products archive organization.

**Figure 6 Chandrayaan-2 Mission Level Archive Organization**

6.4 Instruments Collection

Under each instrument wise data collection, data is organized under each directory defined by PDS4. Data is organized based on the levels of data processing and data products definitions.

To handle requirement of archiving the data products for 12 orbits in a day, data products is organized into year, month and day wise as yyyyymmdd.

The year month and day wise directory contains PDS4 data products including PDS4 Label products. Figures 8 shows the detail of TMC2 instrument wise collection structure.

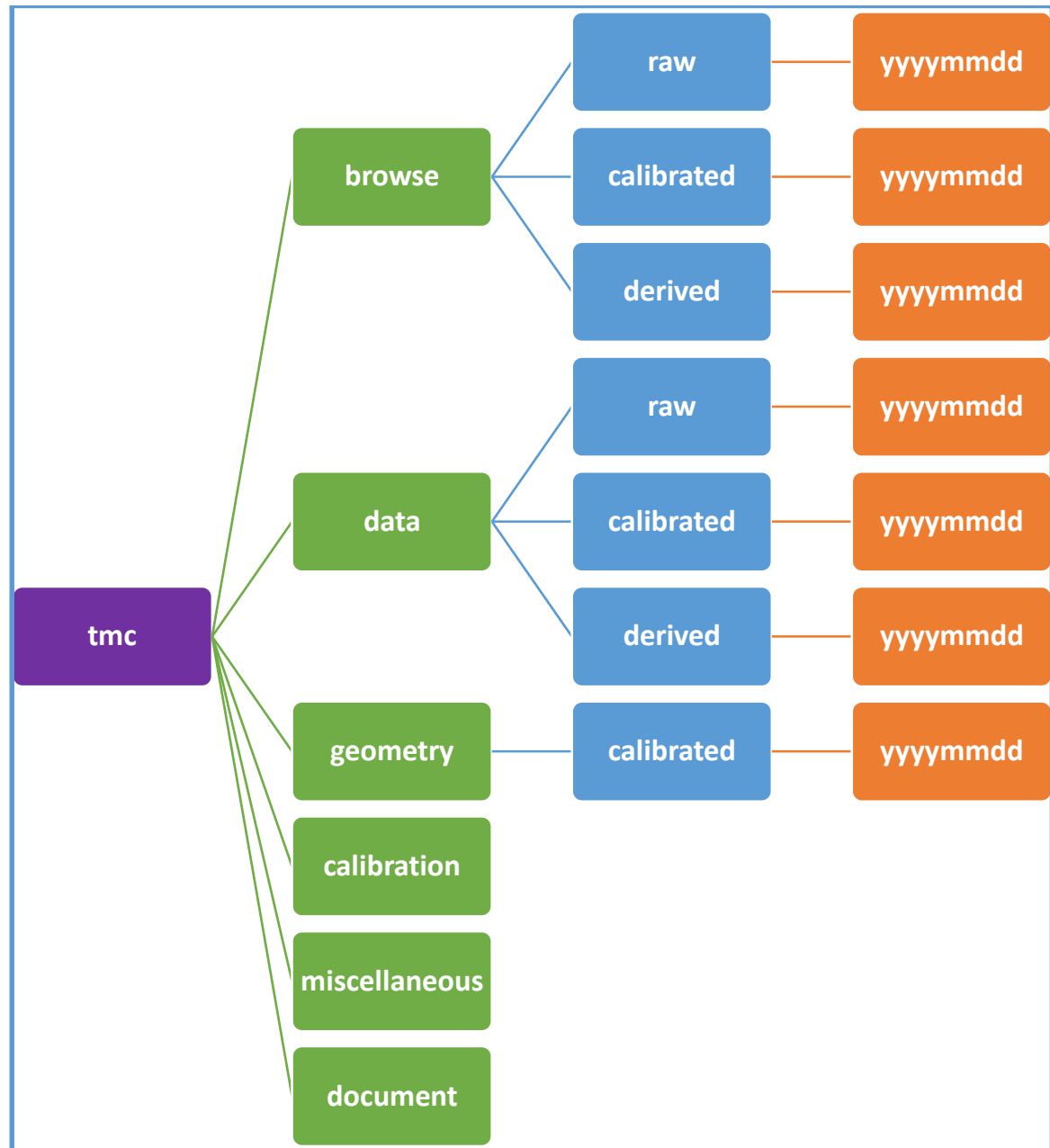


Figure 7 Chandrayaan-2 TMC Instrument wise collection structure

At instrument level, collection of different product types is defined as per standard PDS4 conventions. Table 19 defined in detail about collection types.

Table 18 Collection product types

File or Directory Name	Description
root	Root directory describing full tmc2 collection data sets
browse	Contains 'quick-look' products designed to facilitate use of the archive
calibration	Contains data and files necessary for the calibration of basic products.
context	Contains list of products comprising various objects, identified with in the PDS4 registry, that are specific to the science bundle. These include physical objects such as instruments, spacecraft, and planets and conceptual objects such as missions.
document	Contains all components (figure, tables, text, etc.) of one or more related documents. The document include software interface specifications (SISs), calibration reports.
data	Contains observational products, often separated according to processing level, target, instrument mode, etc.
xml_schema	Contains XML schemas and related products which may be used for generating and validating PDS4 labels.
geometry	Contains non-SPICE geometry products
miscellaneous	Contains supplementary information provided by data provider to be useful in the interpretation and use of other collections in the bundle but which does not fit within the scope of the other collections. for example meta data catalogs, database dumps, and records of modification history could be included.

6.4.1 Root directory content

Files in the root directory include an overview of the archive, a description of the collection for the PDS4 data set. The following files are contained in the Root Directory shown in Table 20.

Table 19 Root directory content products

File Name	File Contents	File Provided By
bundle.xml	Collection content and format information	Archive Team

6.4.2 Browse directory content

Under the Browse directory there is a separate subdirectory which is year, month and day wise for each type of product. The year, month and day wise subdirectory contains browse image for each image, which is a down sampled version of the full resolution image, with the size of m pixels x n lines and 8 bits per pixel, is defined as an object and contained in the data file. PNG images derived from the browse image are stored in this directory, as well as the label files pointing to these images. All the browse PNG images are named as per file naming convention.

6.4.3 Calibration directory content

The calibration directory contains calibration files used to process the data products, or calibration data needed to use the data products. This calibration files are the look up table files that got generated during ground calibration exercise. All these lookup table files naming convention is based on the parameters – mission (ch2), instrument (tmc), sensors (fore, aft and nadir), gain (g1), exposure (e1, e2, e3, e4), version (v1, v2....) and byte selection (msb, lsb and mid). The following files are contained in the calibration directory shown in table 21.

Table 20 Calibration directory content products

File Name	File Contents	File Provided By
collection_calibration_inventory.csv	Contains about the list of calibration data products in the collection inventory	Archive Team
collection_calibration_inventory.xml	XML Label defining the inventory csv file	Archive Team

6.4.4 Context directory content

The context directory contains the list of products comprising various objects, identified with in the PDS4 registry, that are specific to the science bundle. These include physical objects such as instruments, spacecraft, and planets and conceptual objects such as missions. The following files are contained in the context directory shown in table 22.

Table 21 Context directory content products

File Name	File Contents	File Provided By
collection_tmc_context_inventory.csv	Contains about the list of context data products in the collection inventory	Archive Team
collection_tmc_context_inventory.xml	XML Label defining the inventory csv file	Archive Team

6.4.5 Document directory content

The document directory contains documentation to help the user understand and use the archive data. The following files are contained in the document directory shown in the table 23.

Table 22 Document directory content products

File Name	File Contents	File Provided By
collection_document_inventory.xml	Label to define the inventory	Archive Team
collection_document_inventory.csv	A description of the contents of this directory	Archive Team
ch2_tmc_pds_dp_archive_sis.pdf	Description of instrument and PDS data products and Archive Structure	Archive Team
ch2_tmc_pds_dp_archive_sis.xml	XML label defining tmc pds sis pdf	Archive Team

6.4.6 Data directory content

Under the data directory there are three separate subdirectory based on the defined PDS4 data products levels – raw, calibrated and derived. Under each sub directories, there are again sub directory based on year, month and day wise. The year, month and day wise subdirectory contains PDS4 data products mainly image for three sensors defined as Fore, Aft and Nadir and associated label files in XML format. All the browse binary images are named as per file naming conventions mentioned under section 6.5.

6.4.7 Xml_schema directory content

This directory contains XML schemas and related products which may be used for generating and validating PDS4 labels. The following files are contained in the xml_schema directory shown in table 24.

Table 23 Xml_schema directory content products

File Name	File Contents	File Provided By
collection_xml_schema.csv	Contains about the list of xml schema in the collection inventory	Archive Team

collection_xml_schema.xml	XML Label defining the xml schema csv file	Archive Team
---------------------------	---	--------------

6.4.8 Geometry directory content

Under the geometry directory there are three separate subdirectory based on the defined PDS4 data products levels – raw, calibrated and derived. Under each sub directories, there are again sub directory based on year, month and day wise. The year, month and day wise subdirectory contains PDS4 data products mainly grid file in csv format for three sensors defined as Fore, Aft and Nadir and associated label files in XML format. All the grid files are named as per file naming conventions in section 6.5.

6.4.9 Miscellaneous directory content

Under the miscellaneous directory there are two separate subdirectory based on the defined PDS4 data products levels – raw and calibrated. Under each sub directories, there are again sub directory based on year, month and day wise. The year, month and day wise subdirectory contains data products mainly orbit and attitude file (oat), orbit and attitude header file (oath), liberation angle file (lbr) and sun parameter file (spm) in ASCII text format All the files with extension oat, oath, lbr and spm files are named as per file naming conventions in section 6.5. It also contains the supplementary information provided by data provider to be useful in the interpretation and use of other collections in the bundle but which does not fit within the scope of the other collections. The following files are contained in the Document Directory shown in table 25.

Table 24 Miscellaneous directory content

File Name	File Contents	File Provided By
collection_tmc_miscellaneous.csv	Contains about the list of miscellaneous in the collection inventory	Archive Team
collection_tmc_miscellaneous.xml	XML Label defining the xml schema csv file	Archive Team

6.5 File Naming Conventions & Formats

This section describes the naming convention used for science data files for the Chandrayaan-2 mission. Table 25 shows the detail description about the following file naming conventions designed for the data products.

ch2_<inst>_<mtc>_<YYYYMMDDTHHMMSSsss>_<p>_<prd>_<stn>.fff

Table 25 File Naming Conventions

Code	Description
ch2	It denotes the Mission name whose value is fixed as ch2
inst	It denotes the Instrument id, it can have following value = tmc (refers to the TMC2) = iir (refers to the IIRS) = ohr (refers to the OHRC)
m	It denotes the mission phase name, it can have following value = e (refers to the Earth Bound Phase - ebp) = n (refers to the Normal Operations Phase - nop)
t	It denotes the data type; it can have following values: = r (it denotes raw data) = c (It denotes calibrated data) = d (It denotes derived data)
c	It denotes the imaging mode/camera id/band. It can be one of the following: For tmc2 = f (It denotes the fore camera) = a (It denotes the aft camera) = n (it denotes the nadir camera)
YYYYMMDDTHHMMSSsss	It denotes the observation/imaging start time in format [YYYYMMDDTHHMMSSsss] YYYY = stand for year MM = stand for month in MM format DD = stand for day in DD format T = stand for time. HH = stand for hour in HH format MM = stand for minute in MM format SSS = stand for seconds in SS format sss = stand for 1/10 millisecond or 100 microsecond in ssss format

Code	Description
p	It denotes the PDS Data Products categories. It value can be any of the following: = d (It denotes data products under data directory) = b (it denotes browse data products under browse directory) = g (it denotes gridded data products under geometry directory)
prd	It denotes the PDS Data Product Name. It value can be any of the following: = img (It denotes the image data product) = brw (It denotes the browse data product) = grd (It denotes the gridded data product) = dtm (it denotes DTM derived product) = orh (it denotes Ortho derived product)
stn	It denotes the station id from which data was acquired. It value can be any of the following: = d32 (ISSDC Bangalore) = d18 (ISSDC Bangalore) = gds (Gold Stone USA) = cnb (Canberra, Australia)
fff	It denotes the standard file type extension. It can have any one of the following values: = img (PDS Image Data File) = xml (PDS Detached Label File) = jpg (PDS Browse Data File) = csv (PDS Geometry Data File) = tif (PDS Geo tiff data for derived products)

7. Archive Products formats

Data that comprises the TMC 2 archives are formatted in accordance with PDS specifications. This section provides details on the formats used for each of the products included in the archive. The detailed file formats for the defined data products based on PDS4 data types have been defined shown in next sub section.

7.1 Data File Formats

TMC 2 instrument is an imaging payload. The primary data collected from instrument is raw image. The TMC 2 imager has three sensors – Fore, Aft and Nadir. For each sensor both primary as well as secondary data sets are defined that contains data product levels, types and formats shown in the table 27.

Table 26 TMC2 PDS4 Data Products

Sensors	PDS4 Data Product Levels	PDS4 Data Products Types	Data Formats	Data type*
For each sensor - Fore, Aft and Nadir	Raw	Image	Binary	UnsignedLSB2
	Raw	Browse	PNG	UnsignedByte
	Calibrated	Image	Binary	UnsignedLSB2
	Calibrated	Browse	PNG	UnsignedByte
	Calibrated	Geometry	CSV	ASCII Text
Stereo / Triplet Combination	Derived	DEM	Geo Tiff	SignedLSB2 (Float/ double)
	Derived	Ortho	Geo Tiff	UnsignedLSB2
	Derived	Browse DEM	PNG	UnsignedByte
	Derived	Browse Ortho	PNG	UnsignedByte

Note*:

UnsignedLSB2 is UInt16 (Unsigned Integer of 16-bit length)

SignedLSB2 is Int16 (Signed Integer of 16-bit length)

Unsigned Byte is 8 bit

7.2 Document Product File Formats

All the files under document collection will be written in text format, csv, png, xml and pdf

7.3 PDS Labels

PDS labels are ASCII text files written, in the eXtensible Markup Language (XML). All product labels are detached from the digital files (if any) containing the data objects they describe (except Product_Bundle). There is one label for every product. Each product, however, may contain one more data objects. The data objects of a given product may all reside in a single file, or they may be stored in multiple separate files. PDS4 label files must end with the file extension “.xml”.

7.3.1 XML Documents

For the Chandrayaan-2 mission PDS labels will conform to the PDS master based schema based upon the 1.11.0.0 version of the PDS Information Model for structure, and the 1.11.0.0 version of the PDS schematron for content. By use of an XML editor these documents may be used to validate the structure and content of the product labels.

The PDS master schema and schematron documents are produced, managed, and supplied to Chandrayaan-2 by the ISDA Archive Working Group. In addition to these documents, the Chandrayaan-2 Archive Working Group has produced Chandrayaan-2 Archive conventions containing the additional XML attribute and parameter definitions specific to Chandrayaan-2 Orbiter mission.

Examples of PDS labels required for the TMC 2 archive are shown in Appendix B (bundle products), Appendix C (collection products), Appendix D (basic products), Appendix E (Visualization by PDS4 Viewer), Appendix F (Count to Radiance Conversion) and Appendix G (Local Data Dictionary)

7.4 Delivery Package

Data transfers, from ISDA to the data users, are accomplished using delivery packages. Delivery packages include the following required elements:

- The package which consists of a compressed bundle of the products being transferred.

- A transfer manifest which maps each product's LID/VID to the physical location of the product label in the package after uncompression.
- A checksum manifest which lists the MD5 checksum of each file included in the package after uncompression.

Appendix A Support staff and cognizant Persons

Table 27 Chandrayaan-2 Team staff

Teams	Person Name	Contact	Email Id
Operations Team @ ISSDC IDSN Bylalu, Bengaluru	Sri. Sreenath	080-2809-4416/17/18 (SCC), 2809-4419 (MOX), 2202-9173/74 (ISSDC-IDSN)	sreenath@istrac.gov.in
Principal Investigator (PI) Team @ SAC Ahmedabad	DR. A S Arya	07926914107	arya_as@sac.isro.gov.in

Appendix B Sample Bundle Product Label

```

<Product_Bundle xmlns="http://pds.nasa.gov/pds4/pds/v1"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://pds.nasa.gov/pds4/pds/v1
https://pds.nasa.gov/pds4/pds/v1/PDS4_PDS_1700.xsd">
  <Identification_Area>
    <logical_identifier>urn:isro:isda:ch2_cho</logical_identifier>
    <version_id>1.0</version_id>
    <title>Chandrayaan-2 Data Archive Products Bundle</title>
    <information_model_version>1.11.0.0</information_model_version>
    <product_class>Product_Bundle</product_class>
    <Citation_Information>
      <author_list>isro,issdc,sac </author_list>
      <publication_year>2020</publication_year>
      <description>Chandrayaan-2 Orbiter Payloads Data Products</description>
    </Citation_Information>
    <Modification_History>
      <Modification_Detail>
        <modification_date>2020-11-12</modification_date>
        <version_id>1.0</version_id>
        <description>Peer reviewed version of the Chandrayaan-2 Orbiter Payloads data
products bundle</description>
      </Modification_Detail>
    </Modification_History>
  </Identification_Area>
  <Context_Area>
    <Time_Coordinates>
      <start_date_time>2019-11-25Z</start_date_time>
      <stop_date_time>2020-02-12Z</stop_date_time>
    </Time_Coordinates>
    <Investigation_Area>
      <name>Chandrayaan-2</name>
      <type>Mission</type>
      <Internal_Reference>
        <lid_reference>urn:isro:isda:context:investigation:mission.chandrayaan2</lid_refere
nce>
          <reference_type>bundle_to_investigation</reference_type>
        </Internal_Reference>
      </Investigation_Area>
    <Observing_System>
      <name>Chandrayaan-2</name>
      <Observing_System_Component>
        <name>Chandrayaan-2 Orbiter</name>
        <type>Spacecraft</type>
      </Internal_Reference>
    </Observing_System>
  </Context_Area>
</Product_Bundle>

```

```

<lid_reference>urn:isro:isda:context:instrument_host:spacecraft.cho</lid_reference>
>
  <reference_type>is_instrument_host</reference_type>
</Internal_Reference>
</Observing_System_Component>
<Observing_System_Component>
  <name>terrain mapping camera</name>
  <type>Instrument</type>
  <Internal_Reference>
    <lid_reference>urn:isro:isda:context:instrument:tmc.cho</lid_reference>
    <reference_type>is_instrument</reference_type>
  </Internal_Reference>
</Observing_System_Component>
</Observing_System>
<Target_Identification>
  <name>Moon</name>
  <type>Satellite</type>
  <Internal_Reference>
    <lid_reference>urn:isro:isda:context:target:satellite.moon</lid_reference>
    <reference_type>data_to_target</reference_type>
  </Internal_Reference>
</Target_Identification>
</Context_Area>
<Bundle>
  <bundle_type>Archive</bundle_type>
</Bundle>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch2_cho.tmc_collection:calibration</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_calibration_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch2_cho.tmc_collection:data</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_data_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch2_cho.tmc_collection:document</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_document_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch2_cho.tmc_collection:context</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_context_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>

```

```
<lid_reference>urn:isro:isda:ch2_cho.tmc_collection:browse</lid_reference>
<member_status>Primary</member_status>
<reference_type>bundle_has_browse_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch2_cho.tmc_collection:geometry</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_geometry_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>

<lid_reference>urn:isro:isda:ch2_cho.tmc_collection:miscellaneous</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_member_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch2_cho.tmc_collection:schema</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_schema_collection</reference_type>
</Bundle_Member_Entry>
</Product_Bundle>
```

Appendix C Sample Collection Product Label

```
<Product_Collection xmlns="http://pds.nasa.gov/pds4/pds/v1"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://pds.nasa.gov/pds4/pds/v1
http://pds.nasa.gov/pds4/pds/v1/PDS4_PDS_1400.xsd
http://pds.nasa.gov/pds4/disp/v1
http://pds.nasa.gov/pds4/disp/v1/PDS4_DISP_1301.xsd
http://pds.nasa.gov/pds4/sp/v1
http://pds.nasa.gov/pds4/sp/v1/PDS4_SP_1100.xsd">

  <Identification_Area>

    <logical_identifier>urn:isro:isda:ch2_cho.tmc_collection:calibration</logical_identifier>

    <version_id>1.0</version_id>
    <title>Chandrayaan-2 Orbiter TMC 2 calibration file collection</title>
    <information_model_version>1.11.0.0</information_model_version>
    <product_class>Product_Collection</product_class>
    <Citation_Information>
      <author_list>Chandrayaan-2 DP Team</author_list>
      <editor_list>Chandrayaan-2 DP Team</editor_list>
      <publication_year>2020</publication_year>
      <description>
```

This collection contains the calibration files used to calibrate and correct the images acquired by the TMC2 instrument onboard Chandrayaan-2 orbiter spacecraft.

```
    </description>
  </Citation_Information>
  <Modification_History>
    <Modification_Detail>
      <modification_date>2020-11-12</modification_date>
      <version_id>1.0</version_id>
      <description>Initial calibration files</description>
    </Modification_Detail>
  </Modification_History>
</Identification_Area>
<Context_Area>
  <Time_Coordinates>
    <start_date_time>2020-11-12Z</start_date_time>
    <stop_date_time nilReason="inapplicable" xsi:nil="true" />
  </Time_Coordinates>
  <Primary_Result_Summary>
    <purpose>Calibration</purpose>
    <processing_level>Calibrated</processing_level>
  </Primary_Result_Summary>
  <Investigation_Area>
    <name>chandrayaan-2</name>
```

<type>Mission</type>

<Internal_Reference>

<lidvid_reference>urn:isro:isda:context:investigation:mission.chandrayaan2::1.0</lidvid_reference>

<reference_type>collection_to_investigation</reference_type>

</Internal_Reference>

</Investigation_Area>

<Observing_System>

<Observing_System_Component>

<name>Chandrayaan 2 Orbiter</name>

<type>Spacecraft</type>

<description>

Chandrayaan-2 Orbiter is an Orbiter craft under the Chandrayaan-2 Spacecraft consists of various scientific instruments.

</description>

</Observing_System_Component>

<Observing_System_Component>

<name>terrain mapping camera</name>

<type>Instrument</type>

<description>

TMC-2 (Terrain Mapping Camera) is one of the scientific instrument hosted on the Chandrayaan-2 Orbiter. The instrument is passive electro-optical imaging camera operating in visible-panchromatic band (500-800nm), comprising three optical sensors Fore, Nadir and Aft placed with angles of +25 deg, 0 deg and -25 deg to have three stereo views of the same target as spacecraft moves along the track.

</description>

</Observing_System_Component>

</Observing_System>

<Target_Identification>

<name>Moon</name>

<type>Satellite</type>

<description>Moon is a natural satellite of Earth</description>

</Target_Identification>

</Context_Area>

<Reference_List>

<Internal_Reference>

<lid_reference>urn:isro:isda:ch2_cho.tmc_collection:document:ch2missioninfo</lid_reference>

<reference_type>collection_to_document</reference_type>

</Internal_Reference>

<Internal_Reference>

<lid_reference>urn:isro:isda:ch2_cho.tmc_collection:document:tmc2dpsis</lid_reference>

```

    <reference_type>collection_to_document</reference_type>
  </Internal_Reference>
</Reference_List>
<Collection>
  <collection_type>Calibration</collection_type>
</Collection>
<File_Area_Inventory>
  <File>
    <file_name>collection_calibration_inventory.csv</file_name>
    <local_identifier>collection_calibration_inventory.csv</local_identifier>
    <creation_date_time>2020-07-20</creation_date_time>
    <file_size unit="byte">2730</file_size>
    <records>39</records>
    <md5_checksum>2b7e0e0565579b8c57f07c75065f0d17</md5_checksum>
  </File>
  <Inventory>
    <offset unit="byte">0</offset>
    <parsing_standard_id>PDS DSV 1</parsing_standard_id>
    <records>39</records>
    <record_delimiter>Carriage-Return Line-Feed</record_delimiter>
    <field_delimiter>Comma</field_delimiter>
    <Record_Delimited>
      <fields>2</fields>
      <groups>0</groups>
      <maximum_record_length unit="byte">259</maximum_record_length>
      <Field_Delimited>
        <name>Member Status</name>
        <field_number>1</field_number>
        <data_type>ASCII_String</data_type>
        <maximum_field_length unit="byte">1</maximum_field_length>
        <description>This column specifies the member status</description>
      </Field_Delimited>
      <Field_Delimited>
        <name>LIDVID_LID</name>
        <field_number>2</field_number>
        <data_type>ASCII_LIDVID_LID</data_type>
        <maximum_field_length unit="byte">255</maximum_field_length>
        <description>This column specifies the LIDVID of the files that comprise the
collection. </description>
      </Field_Delimited>
    </Record_Delimited>
    <reference_type>inventory_has_member_product</reference_type>
  </Inventory>
</File_Area_Inventory>

</Product_Collection>

```

Appendix D Sample Data Product Labels

```

<Product_Observational xmlns="http://pds.nasa.gov/pds4/pds/v1"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://pds.nasa.gov/pds4/pds/v1
http://pds.nasa.gov/pds4/pds/v1/PDS4_PDS_1400.xsd
http://pds.nasa.gov/pds4/disp/v1
http://pds.nasa.gov/pds4/disp/v1/PDS4_DISP_1301.xsd
http://pds.nasa.gov/pds4/sp/v1
http://pds.nasa.gov/pds4/sp/v1/PDS4_SP_1100.xsd">
  <Identification_Area>
    <logical_idenfier>urn:isro:isda:ch2_cho:tmc_calibrated:ch2_tmc_ncn_20200131t10
10006367_d_img_d18</logical_idenfier>
    <version_id>1.0</version_id>
    <title>Chandrayaan-2 Orbiter TMC2 Experiment</title>
    <information_model_version>1.11.0.0</information_model_version>
    <product_class>Product_Observational</product_class>
    <Modification_History>
      <Modification_Detail>
        <modification_date>2019-05-07</modification_date>
        <version_id>1.0</version_id>
        <description>PDS4 product label created by SAC Optical DP Team</description>
      </Modification_Detail>
    </Modification_History>
  </Identification_Area>
  <Observation_Area>
    <Time_Coordinates>
      <start_date_time>2020-01-31T10:10:00.6367Z</start_date_time>
      <stop_date_time>2020-01-31T10:19:44.4410Z</stop_date_time>
    </Time_Coordinates>
    <Primary_Result_Summary>
      <purpose>Science</purpose>
      <processing_level>Calibrated</processing_level>
    </Primary_Result_Summary>
    <Investigation_Area>
      <name>chandrayaan-2</name>
      <type>Mission</type>
      <Internal_Reference>
        <lidvid_reference>urn:isro:isda:context:investigation:mission.chandrayaan2::1.0</lid
vid_reference>
        <reference_type>data_to_investigation</reference_type>
      </Internal_Reference>
    </Investigation_Area>
    <Observing_System>
      <Observing_System_Component>
        <name>Chandrayaan 2 Orbiter</name>
        <type>Spacecraft</type>
        <description>

```

Chandrayaan-2 Orbiter is an Orbiter craft under the Chandrayaan-2 Spacecraft consists of various scientific instruments.

```

</description>
</Observing_System_Component>
<Observing_System_Component>
  <name>terrain mapping camera</name>
  <type>Instrument</type>
  <description>
    TMC-2 (Terrain Mapping Camera) is one of the scientific instrument hosted
    on the Chandrayaan-2 Orbiter. The instrument is passive electro-optical
    imaging camera operating in visible-panchromatic band (500-800nm),
    comprising three optical sensors Fore, Nadir and Aft placed with angles
    of -25 deg, 0 deg and +25 deg to have three stereo views of the same target
    as spacecraft moves along the track.
  </description>
</Observing_System_Component>
</Observing_System>
<Target_Identification>
  <name>Moon</name>
  <type>Satellite</type>
  <description>Moon is a natural satellite of Earth</description>
</Target_Identification>
<Mission_Area>
  <Product_Parameters>
    <job_id>TMCXXD18CHO0194500NNNN20031114535386_V1_0</job_id>
<level0_dir_name>TMCXXD18CHO0194500NNNN20031114535386_V1_0</level0_dir_name>
    <imaging_orbit_number>1945</imaging_orbit_number>
    <dumping_orbit_number>1945</dumping_orbit_number>
    <line_exposure_duration unit="ms">3.236</line_exposure_duration>
    <gain>g1</gain>
    <exposure>e1</exposure>
    <detector_pixel_width unit="micron">7</detector_pixel_width>
    <focal_length unit="mm">140</focal_length>
    <reference_data_used>SELENE</reference_data_used>
    <orbit_limb_direction>Ascending</orbit_limb_direction>
    <spacecraft_yaw_direction>True</spacecraft_yaw_direction>
    <spacecraft_altitude unit="km">83.027613236360025</spacecraft_altitude>
  </Product_Parameters>
  <Geometry_Parameters>
    <System_Level_Coordinates>
      <upper_left_latitude unit="deg">-32.025187</upper_left_latitude>
      <upper_left_longitude unit="deg">69.419255</upper_left_longitude>
      <upper_right_latitude unit="deg">-32.013041</upper_right_latitude>
      <upper_right_longitude unit="deg">68.758781</upper_right_longitude>
      <lower_left_latitude unit="deg">-1.743489</lower_left_latitude>
      <lower_left_longitude unit="deg">69.934485</lower_left_longitude>
    </System_Level_Coordinates>
  </Geometry_Parameters>
</Mission_Area>

```

```

    <lower_right_latitude unit="deg">-1.733504</lower_right_latitude>
    <lower_right_longitude unit="deg">69.386512</lower_right_longitude>
  </System_Level_Coordinates>
  <Refined_Corner_Coordinates>
    <upper_left_latitude unit="deg">-32.054787</upper_left_latitude>
    <upper_left_longitude unit="deg">69.454792</upper_left_longitude>
    <upper_right_latitude unit="deg">-32.042210</upper_right_latitude>
    <upper_right_longitude unit="deg">68.783917</upper_right_longitude>
    <lower_left_latitude unit="deg">-1.781664</lower_left_latitude>
    <lower_left_longitude unit="deg">69.953822</lower_left_longitude>
    <lower_right_latitude unit="deg">-1.772598</lower_right_latitude>
    <lower_right_longitude unit="deg">69.397184</lower_right_longitude>
  </Refined_Corner_Coordinates>
</Geometry_Parameters>
</Mission_Area>
</Observation_Area>
<File_Area_Observational>
  <File>
    <file_name>ch2_tmc_ncn_20200131T1010006367_d_img_d18.img</file_name>
    <creation_date_time>2020-07-07T18:04:13</creation_date_time>
    <file_size unit="byte">1443272000</file_size>
    <md5_checksum>9820cdde2c9870bead1926243dbe06a8</md5_checksum>
    <comment>

```

This File contains the count calibrated image. md5_checksum is provided for ensuring data integrity when users are downloading the data.

```

    </comment>
  </File>
  <Array_2D_Image>
    <offset unit="byte">0</offset>
    <axes>2</axes>
    <axis_index_order>Last Index Fastest</axis_index_order>
    <Element_Array>
      <data_type>UnsignedLSB2</data_type>
    </Element_Array>
    <Axis_Array>
      <axis_name>Line</axis_name>
      <elements>180409</elements>
      <sequence_number>1</sequence_number>
    </Axis_Array>
    <Axis_Array>
      <axis_name>Sample</axis_name>
      <elements>4000</elements>
      <sequence_number>2</sequence_number>
    </Axis_Array>
  </Array_2D_Image>
</File_Area_Observational>
</Product_Observational>

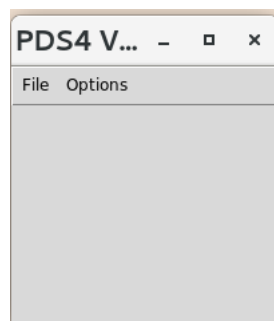
```

Appendix E Visualize PDS4 Data Products using PDS4 Viewer

There are many PDS tools are available for opening the Raw/Calibrated Data products. PDS4Viewer - PDS4Viewer is a simple display tool for visualizing data contained in the PDS4 data standard. PDS4Viewer tool can be downloaded from internet using link “http://sbndev.astro.umd.edu/wiki/PDS4_Viewer”. This Software is available for both Linux as well as window platform. The given data sets can be opened using PDS4Viewer.

The instructions to open the data using PDS4Viewer are as follows:

1. Download the PDS4Viewer tool from internet http://sbndev.astro.umd.edu/wiki/PDS4_Viewer. There will be a zip file (for both Linux/Windows) contains the PDS4Viewer.
2. Extract the Zip file into any location.
3. Go to extract the directory, run pds4_viewer. The PDS4Viewer GUI will come which is shown below.



Opening PDS4 Raw/Calibrated Image Product

=====

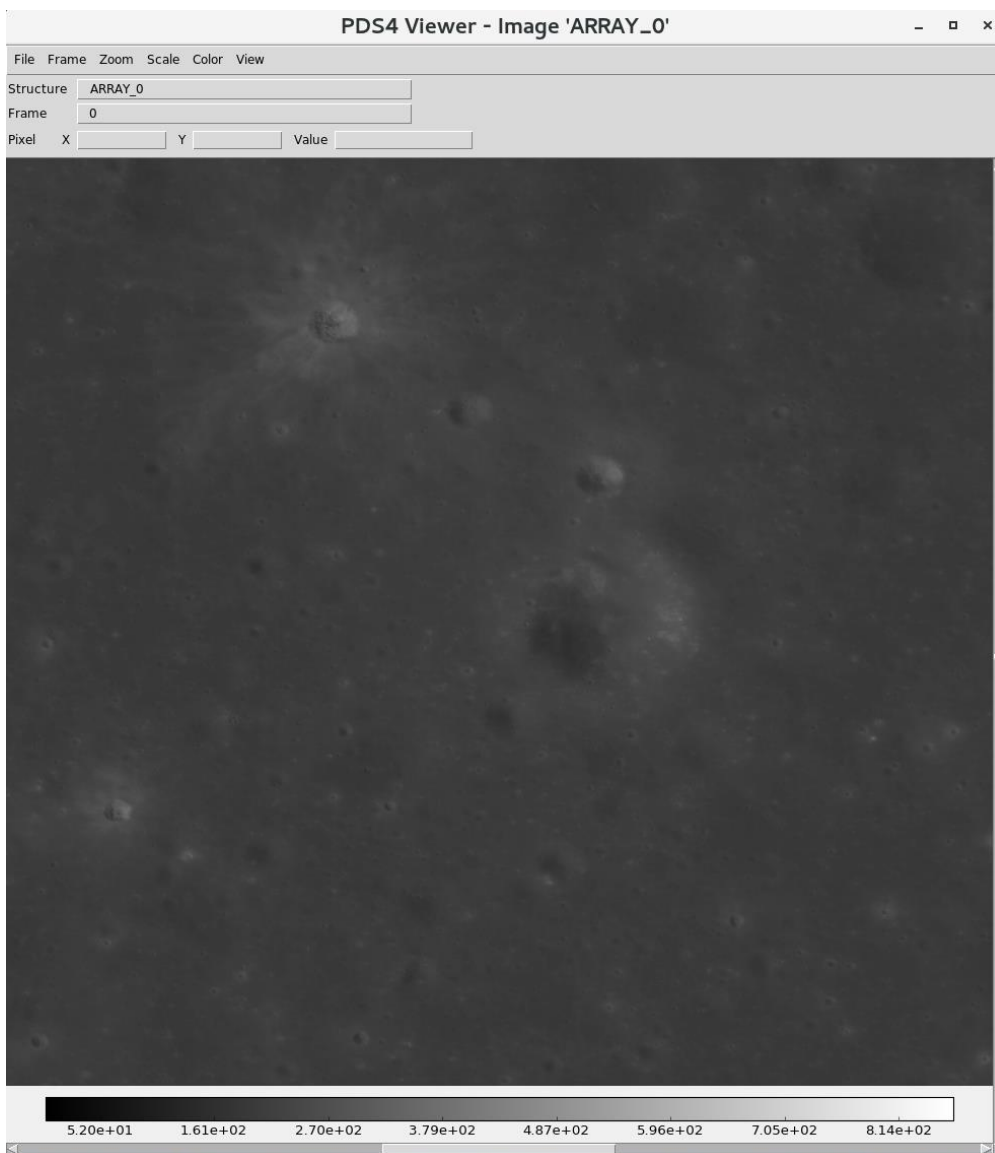
1. After opening the PDS4Viewer display tool. Click on the File menu and select Open Object
2. Click file open dialog will open which is shown below select Label File and click Open button



3. After clicking Open button, the data will be loaded in the PDS4Viewer shown below:



4. For Opening an image, click on the Image button shown on the above figure at step no. 3. Below figure shows the processed calibrated image acquired by TMC Nadir Camera



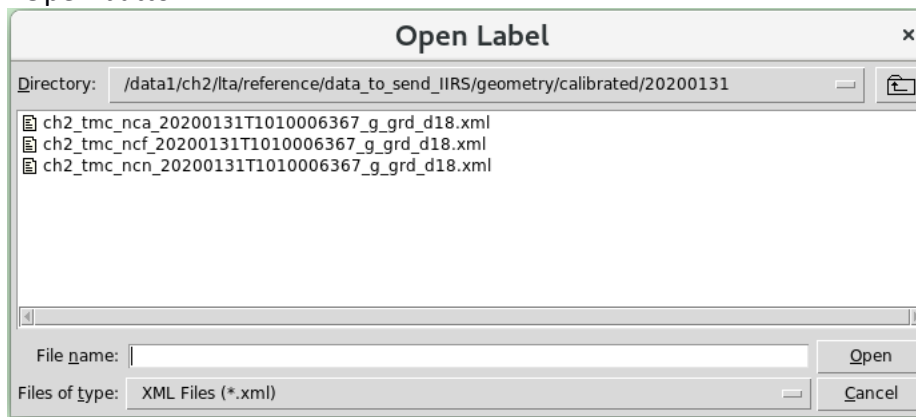
5. For Opening a label, click on the Label by selecting from View drop down menu. Below figures shows the label file as label view and xml view

PDS4 Viewer - Label View	
File Edit View	
Label	
Product_Observational	
Identification Area	
<u>logical_identifier</u> : urn:isro:isda:ch2_cho:tmc_calibrated:ch2_tmc_ncn_20200131t1010006367_d_img_d18 <u>version_id</u> : 1.0 <u>title</u> : Chandrayaan-2 Orbiter TMC2 Experiment <u>information_model_version</u> : 1.11.0.0 <u>product_class</u> : Product_Observational	
Modification History	
Modification Detail	
<u>modification_date</u> : 2019-05-07 <u>version_id</u> : 1.0 <u>description</u> : PDS4 product label created by SAC Optical DP Team	
Observation Area	
Time Coordinates	
<u>start_date_time</u> : 2020-01-31T10:10:00.6367Z <u>stop_date_time</u> : 2020-01-31T10:19:44.4410Z	
Primary Result Summary	
<u>purpose</u> : Science <u>processing_level</u> : Calibrated	
Investigation Area	
<u>name</u> : chandrayaan-2 <u>type</u> : Mission	
Internal Reference	
<u>lvidid_reference</u> : urn:isro:isda:context:investigation:mission.chandrayaan2::1.0 <u>reference_type</u> : data_to_investigation	
Observing System	
Observing System Component	
<u>name</u> : Chandrayaan 2 Orbiter <u>type</u> : Spacecraft <u>description</u> : Chandrayaan-2 Orbiter is an Orbiter craft under the Chandrayaan-2 Spacecraft consists of various scientific instruments.	
Observing System Component	
<u>name</u> : terrain mapping camera <u>type</u> : Instrument <u>description</u> : TMC-2 (Terrain Mapping Camera) is one of the scientific instrument hosted on the Chandrayaan-2 Orbiter. The instrument is passive electro-optical imaging camera operating in visible-panchromatic band (500-800nm), comprising three optical sensors Fore, Nadir and Aft placed with angles of -25 deg, 0 deg and +25 deg to have three stereo views of the same target as spacecraft moves along the track.	
Target Identification	
<u>name</u> : Moon <u>type</u> : Satellite <u>description</u> : Moon is a natural satellite of Earth	
Mission Area	
Product Parameters	
<u>job_id</u> : TMCXXD18CH00194500NNNN20031114535386_V1_0 <u>level0_dir_name</u> : TMCXXD18CH00194500NNNN20031114535386_V1_0 <u>imaging_orbit_number</u> : 1945 <u>dumping_orbit_number</u> : 1945 <u>line_exposure_duration</u> : 3.236 <u>gain</u> : g1 <u>exposure</u> : e1 <u>detector_pixel_width</u> : 7 <u>focal_length</u> : 140 <u>reference_data_used</u> : SELENE <u>orbit_limb_direction</u> : Ascending <u>spacecraft_yaw_direction</u> : True <u>spacecraft_altitude</u> : 83.027613236360025	
Geometry Parameters	
System Level Coordinates	
<u>upper_left_latitude</u> : -32.025187 <u>upper_left_longitude</u> : 69.419255 <u>upper_right_latitude</u> : -32.013041 <u>upper_right_longitude</u> : 68.758781 <u>lower_left_latitude</u> : -1.743489 <u>lower_left_longitude</u> : 69.934485	

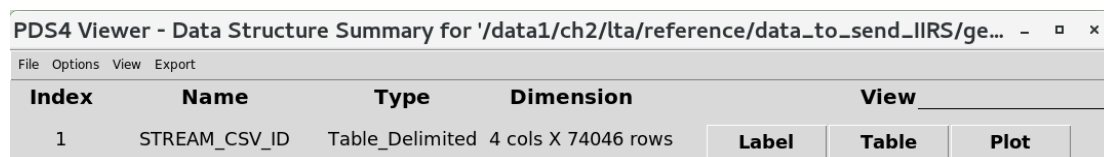
Opening PDS4 Geometry Grid Data Product

=====

1. After opening the PDS4Viewer display tool. Click on the File menu and select Open Object
2. Click file open dialog will open which is shown below select Label File and click Open button



3. After clicking Open button, the data will be loaded in the PDS4Viewer shown below:



4. For Opening grid file, click on the table button shown on the above figure at step no. 6. Below shows the grid file from the TMC2 Payloads

PDS4 Viewer - Table 'STREAM_CSV_ID' - □ ×

File Options View

Row #	Longitude	Latitude	Pixel	Scan
0	0.4547922	-32.0547874	0	0
1	69.4379765	-32.0545145	100	0
2	69.4211636	-32.0542395	200	0
3	69.4043535	-32.0539623	300	0
4	69.3875461	-32.0536831	400	0
5	69.3707414	-32.0534016	500	0
6	69.3539392	-32.0531181	600	0
7	69.3371396	-32.0528324	700	0
8	69.3203424	-32.0525446	800	0
9	69.3035476	-32.0522546	900	0
10	69.2867551	-32.0519625	1000	0
11	69.2699649	-32.0516683	1100	0
12	69.2531769	-32.051372	1200	0
13	69.236391	-32.0510735	1300	0
14	69.2196072	-32.0507729	1400	0
15	69.2028254	-32.0504701	1500	0
16	69.1860456	-32.0501653	1600	0
17	69.1692676	-32.0498582	1700	0
18	69.1524915	-32.0495491	1800	0
19	69.1357171	-32.0492378	1900	0
20	69.1189444	-32.0489244	2000	0
21	69.1021733	-32.0486088	2100	0
22	69.0854039	-32.0482911	2200	0
23	69.0686359	-32.0479713	2300	0
24	69.0518693	-32.0476493	2400	0
25	69.0351042	-32.0473252	2500	0
26	69.0183403	-32.046999	2600	0
27	69.0015777	-32.0466706	2700	0
28	68.9848163	-32.0463401	2800	0
29	68.968056	-32.0460074	2900	0
30	68.9512968	-32.0456726	3000	0
31	68.9345386	-32.0453356	3100	0
32	68.9177814	-32.0449965	3200	0
33	68.901025	-32.0446553	3300	0
34	68.8842694	-32.0443119	3400	0
35	68.8675146	-32.0439663	3500	0
36	68.8507605	-32.0436186	3600	0
37	68.834007	-32.0432688	3700	0
38	68.817254	-32.0429168	3800	0
39	68.8005016	-32.0425627	3900	0
40	68.7839171	-32.0422099	3999	0
41	69.4551062	-32.0380298	0	100
42	69.4382941	-32.037757	100	100

5. For Opening a label, click on the Label by selecting from View drop down menu. Below figures shows the label file as label view and xml view

PDS4 Viewer - Label View

File Edit View

Label

Product_Observational

Identification_Area
logical_identifier: urn:isro:isda:ch2_cho:tmc_calibrated:ch2_tmc_ncn_20200131t1010006367_g_grd_d18
version_id: 1.0
title: Chandrayaan-2 Orbiter TMC 2 Experiment
information_model_version: 1.11.0.0
product_class: Product_Observational

Modification_History

Observation_Area

Time_Coordinates
start_date_time: 2020-01-31T10:10:00.6367Z
stop_date_time: 2020-01-31T10:19:44.4410Z

Investigation_Area
name: Chandrayaan-2
type: Mission

Internal_Reference
lidvid_reference: urn:isro:isda:context:investigation:mission.chandrayaan2::1.0
reference_type: data_to_investigation

Observing_System

Observing_System_Component
name: Chandrayaan 2 Orbiter
type: Spacecraft
description:
Chandrayaan-2 Orbiter is an Orbiter craft under the Chandrayaan-2 Spacecraft consists of various scientific instruments.

Observing_System_Component
name: terrain mapping camera
type: Instrument
description:
TMC-2 (Terrain Mapping Camera) is one of the scientific instrument hosted on the Chandrayaan-2 Orbiter. The instrument is passive electro-optical imaging camera operating in visible-panchromatic band (500-800nm), comprising three optical sensors Fore, Nadir and Aft placed with angles of -25 deg, 0 deg and +25 deg to have three stereo views of the same target as spacecraft moves along the track.

Target_Identification
name: Moon
type: Satellite
description: Moon is a natural satellite of Earth

File_Area_Observational

File
file_name: ch2_tmc_ncn_20200131T1010006367_g_grd_d18.csv
local_identifier: STREAM DELIMITED CSV FILE_ID
creation_date_time: 2020-07-07T18:04:13
file_size: 2581997
records: 74046
md5_checksum: 0ac8377f440702779b7a3a3150b6bb86
comment:
This File contains selenographic coordinates for the image points.
md5_checksum is provided for ensuring data integrity when users are downloading the data.

Header
name: Column headings for Grid Data
local_identifier: HEADER
offset: 0
object_length: 31
parsing_standard_id: PDS DSV 1

Table_Delimited
local_identifier: STREAM_CSV_ID
offset: 33
parsing_standard_id: PDS DSV 1
records: 74046
record_delimiter: Carriage-Return Line-Feed
field_delimiter: Comma

Record_Delimited
fields: 4
groups: 0

Field_Delimited
name: Longitude
field_number: 1
data_type: ASCII_Real

Appendix F Count to Radiance Conversion

The TMC2 Raw and Calibrated image data products are the count data products in order to convert to radiance data product following formula is used for the conversion

1. Formula for Conversion of Count to Radiance

$$\text{radiance value} = \frac{(\text{Max_SR} - \text{Min_SR}) * \text{gray_value}}{\text{no_of_gray_value}}$$

Note:

- SR stands for Saturation Radiance.
- number_of_gray_value will be calculated as 2^n , where n is quantization bit (for tmc2 it is 10 bit).
- Min_SR and Max_SR are the minimum Saturation and maximum Radiances respectively.
- In TMC2, there are single gain i.e. g1 and four exposures e1, e2, e3, e4. For a particular combination of exposure & gain value, we obtained Maximum Saturation Radiance value & Minimum Saturation Radiance Value corresponding to them from a table. As in TMC, there will be three table files depending upon the Fore, Aft, and Nadir camera.

Table 29 shows the saturation radiance information for TMC2 cameras - Fore, Aft and Nadir.

Table 28 TMC2 Saturation Information

TMC2 Camera	Gain	Exposure	Min_SR	Max_SR
Fore	1	1	0	14.41
	1	2	0	16.58
	1	3	0	20.08
	1	4	0	29.04
Aft	1	1	0	14.23
	1	2	0	16.36
	1	3	0	19.81
	1	4	0	28.61
Nadir	1	1	0	13.91
	1	2	0	15.99
	1	3	0	19.38
	1	4	0	28.02

Appendix G Local Data Dictionary

Every mission has some conditions based on the events that needs to be handle as an information at output product level. Some information is common to every planetary mission launched by ISRO. There is an ISRO Science Data Archive established during Chandrayaan-1 mission in 2008. All the products hosted for users under ISDA are PDS compliance. For Chandrayaan-2 mission, PDS4 Archive standard was adopted. PDS4 is currently next and future generation archive standard adopted by all the space agencies including NASA, ESA and JAXA for their current and future mission. To handle the ISRO planetary mission's specific parameters, there is a concept in PDS4 called Local Data Dictionary i.e. LDD. In LDD one can define all the parameters and that can be used as a reference for future planetary missions in India. LDD takes basic data dictionary of PDS4 and then top of it builds local dictionary. For Chandrayaan-2 mission and future planetary missions of ISRO, list of parameters is identified shown in details in table 29.

Table 29 ISDA Local Data Dictionary Class and Attributes List

Class/Sub Class Hierarchy	Attribute Name	Description/Value
Mission_Area Product_Parameters	job_id	Unique id assign by DP Scheduler when the product is ingested
	level0_dir_name	DP Input Level-0 directory name
	imaging_orbit_number	Imaging Orbit Number assigned on board
	dumping_orbit_number	Dumping Orbit Number is assigned based on the data collected on ground station
	line_exposure_duration	Image integration time
	gain	Gain is one of the operating modes. It is value is set as g1
	exposure	exposure is one of the operating modes. It is value is set as e1 or e2
	detector_pixel_width	Pixel width of detector
	focal_length	Instrument focal length
	spacecraft_altitude	Spacecraft height
	orbit_limb_direction	Orbit limb direction it can have two values - Ascending or Descending
	spacecraft_yaw_direction	Spacecraft yaw mode it can have two values – True/False

Class/Sub Class Hierarchy	Attribute Name	Description/Value
		a) True means Reverse, in flight yaw mode Reverse describes 180 degree rotated from nominal. b) False means Forward, in flight yaw mode Forward describes 0 degree rotated from nominal) There are four possible orbit_limb_direction and spacecraft_yaw_direction combinations: 1. Descending and False, no changes in lines or samples are performed 2. Descending and True, samples are reversed 3. Ascending and False, Lines are reversed 4. Ascending and True. both lines and samples are reversed
	reference_data_used	Reference data set used in refining the system level corners List of Possible Values 1. SELENE (10 meter ortho reference) 2. LRO (100 meter ortho reference) 3. CLEMENTINE (100 meter ortho reference) 4. System (available reference but no match found gone with system level) 5. n/a means not available
	pixel_resolution	Image pixel resolution expressed in unit

Class/Sub Class Hierarchy	Attribute Name	Description/Value
		m/pixel
	product_accuracy_stdev_height	Product Quality check accuracy standard deviation height parameter if value is n/a means not available otherwise value expressed in unit meter
	product_accuracy_rmse_height	Product Quality check accuracy RMSE (Root Mean Square Error) parameter if value is n/a means not available otherwise value expressed in unit meter
	product_accuracy_stdev_CE	Product Quality check accuracy standard deviation CE (Circular Error) parameter if value is n/a means not available otherwise value expressed in unit meter
	product_accuracy_rmse_CE	Product Quality check accuracy RMSE (Root Mean Square Error) CE (Circular Error) parameter if value is n/a means not available otherwise value expressed in unit meter
	roll	Spacecraft spinning on X-axis value expressed in degree
	pitch	Spacecraft spinning on Y-axis value expressed in degree
	yaw	Spacecraft spinning on Z-axis value expressed in degree
	sun_azimuth	Angle between the projection of sun rays and a line due south or north and value is expressed in degree
	sun_elevation	Angle between the horizon and the Sun's disc centre and value is

Class/Sub Class Hierarchy	Attribute Name	Description/Value
		expressed in degree
	solar_incidence	Angle between the sun's rays and the normal on a surface and value is expressed in degree
	projection	projection is map projection which is set of transformations employed to represent the curved two-dimensional surface of a globe on plane. List of possible values 1. Selenographic 2. Polar stereographic
	area	Area on which imaging is done. List of Possible Values 1. Equatorial 2. North Pole 3. South Pole
Mission_Area - Geometry_Parameters - System_Level_Coordinates - Refined_Level_Coordinates - Corrected_Level_Coordinates	upper_left_latitude	Upper left latitude reticle point with number 1
	upper_left_longitude	Upper left longitude reticle point with number 1
	upper_right_latitude	Upper right latitude reticle point with number 3
	upper_right_longitude	Upper right longitude reticle point with number 3
	lower_left_latitude	Lower left latitude reticle point with number 7
	lower_left_longitude	Lower left longitude reticle point with number 7
	lower_right_latitude	Lower right latitude reticle point with number 9
	lower_right_longitude	Lower right longitude reticle point with number 9

Appendix H Data Disclaimer

The public release of TMC-2 data products has three levels- Level-0 (Raw), Level-1 (Calibrated) and Level-2 (Derived).

Level-0 data is the uncalibrated data in count numbers (10-bit unsigned integer, generic binary sequential (BSQ) format). It can be converted to level-1 data by using the equation along with the maximum & minimum saturation radiances provided in Appendix F of DPSIS document.

Level-1 is radiometric calibrated data (generic binary format) with seleno tagged information. This data has been generated by using level-0 data and the instrument calibration coefficients (LUT).

Level-2 Data is derived data from TMC-2 Fore, Aft and Nadir Image. This data contains Digital Elevation model (DEM) and Ortho-image of Nadir camera data. These datasets are provided in Geotiff format to allow users more flexibility in using the DEM.

TMC-2 datasets generated DEM may have these limitations:

1. Each orbit is processed independently using the automatic identification of control points from reference images (SELENE, LRO WAC and Clementine global mosaic). As we know that the spatial resolution of each reference images are varying from ~8.5m to ~100m, because of that the planimetry accuracy of each product also will vary depending upon the reference image used. The accuracy variations are in the order of ~10 reference image pixels. The different reference images are used for accounting to illumination difference between TMC and reference images for matching. There are cases where lunar control point couldn't be automatically identified and in those cases products are generated with system level knowledge, which will result in planimetry inaccuracy of ~2.5km. User will also may observe some amount of positional shift between ortho and DEM.
2. The adjacent orbit of TMC are not bundle block adjusted and user may observe positional shift between adjacent orbits. This will be taken after global mosaic generation using all the orbits of TMC datasets.
3. The DEM may contain small artefacts, blunders and failed values at few locations in the entire DEM file. This is because areas that were very bland (low contrast) or deeply shadowed with low contrast and low signal may have a "faceted" look to them. This is due to the processing algorithm not able to correlate the images and must interpolate to fill in the pixel. The terrain data in these areas are not trustworthy.
4. DEMs may also have varying level of noise resulted from low signal images, low incidence angle, and jitter.
5. The position and height accuracies of DEMs may be poorer in few areas because of in-adequacy of lunar control points for perfect geometrical modelling.